

Shuyu Wang

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RESEARCH INTERESTS

My research focuses on the dynamics and evolution of the Milky Way and the Magellanic Clouds, combining numerical simulations with observational data. In my previous work, I used N-body+SPH simulations to analyze the origin of the phase space spirals, a signature of past perturbations to the Milky Way disk. I currently study the morphology and kinematics of the Large Magellanic Cloud using N-body simulations. I compare simulation results with Gaia observational data.

Keywords: Milky Way dynamics, Magellanic Clouds, galactic dynamics, galactic archaeology

EDUCATION

Ph.D. Candidate in Astronomy

Sep 2023 - Sep 2027 (expected)

Leiden Observatory, Leiden University, Leiden, The Netherlands

Supervisor: Anthony Brown

Funding: The Horizon Europe Marie Skłodowska-Curie Actions Doctoral Network *MWGaiaDN*

Thesis theme: The Dynamics of the Milky Way and its Massive Neighbors

Master of Science in Physical Sciences

Sep 2020 - Dec 2021

University of Chicago, Chicago, USA

Supervisor: Alexander Ji

Thesis: [High-resolution Chemical Abundances of the Nyx Stream](#)

Bachelor of Science in Physics

Aug 2016 - Jun 2020

Southern University of Science and Technology, Shenzhen, China

Supervisor: Leonardo Modesto

Thesis: Analysis of Nonlocal Gravity in Terms of Auxiliary Fields

Undergraduate Exchange Student

Aug 2019 - Dec 2019

University of California Berkeley, Berkeley, USA

Project paper: The Overview of Accretion Disk Models around Active Galactic Nuclei (AGN)

PUBLICATIONS

Wang, S., Schölich, M., Romero-Gómez, M., Jiménez-Arranz, Ó., Brown, A.G.A., *Tidally-induced vertical asymmetries in the LMC bar and disk*, (in-prep).

Wang, S., Brown, A.G.A., Debattista, V.P., Khachaturyants, T. 2025, *Phase Spirals Induced by the Gas Warp*, [A&A](#), 710, A120 (2026).

Wang, S., Necib, L., Ji, A.P., et al. 2023, *High-resolution chemical abundances of the Nyx stream*, [ApJ](#), 955 129.

RESEARCH PROJECTS

Tidally-induced vertical asymmetries in the LMC bar and disk 2026

Leiden University & University of Barcelona

Supervisors: M. Romero-Gómez & A.G.A. Brown

Phase spirals induced by the gas warp 2023 - 2025

Leiden University & University of Lancashire

Supervisors: A.G.A. Brown & V.P. Debattista

Sensitivity of the CSST telescope for exoplanet microlensing event detection 2022

Tsinghua University

Supervisor: W. Zhu

High-resolution chemical abundances of the Nyx stream 2021 - 2022

University of Chicago

Supervisors: A.P. Ji & L. Necib

Deep learning basics and strong lensing 2021

University of Chicago

Supervisor: B. Nord

Accretion Disk Models around Active Galactic Nuclei 2019

University of California Berkeley

Supervisor: J. Lu

TECHNICAL SKILLS

- **Scientific programming:** Python, Shell/Bash, SQL, L^AT_EX, Git
- **High-performance and parallel computing:** multiprocessing, JAX, GNU Parallel
- **Galactic dynamics and modelling:** AGAMA, galpy; test particle simulation setup and analysis
- **Numerical simulations:** N-body and N-body+SPH simulation snapshot analysis and processing
- **Topological data analysis:** Applying persistent homology to simulation data using GUDHI
- **Statistical inference:** Markov Chain Monte Carlo (MCMC) using `emcee`
- **Machine learning:** scikit-learn (dimensionality reduction using PCA, clustering using K-Means), TensorFlow (basics)
- **Observational astronomy:**
 - Spectral reduction and processing using SMH
 - 3 (remote) nights on Magellan (MIKE), PI: A.P. Ji
- **Language:** Chinese (native), English (fluent)

Oct 2021

RESEARCH VISITS

- **University of Barcelona**, Barcelona, Spain (hosted by Prof. Mercè Romero-Gómez) Jul 2026
Collaboration on analyzing the vertical asymmetries of the LMC bar and disk using N-body simulations and mock Gaia catalogs
- **University of Lancashire**, Lancashire, UK (hosted by Prof. Victor P. Debattista) Jan & Dec 2024
Collaboration on analyzing the phase space spirals using N-body+SPH simulations

PRESENTATIONS

- **Phase spirals induced by the gas warp** (Poster with short presentation)
European Astronomical Society Annual Meeting (EAS 2026), Lausanne, Switzerland Jun 2026
- **Phase spirals induced by the gas warp** (Poster)
Galactic bars, bulges and disks workshop, Lipari, Italy Jun 2026
- **Phase spirals induced by the gas warp** (Contributed talk)
Nederlandse Astronomen Conferentie (NAC 2026), Egmond aan Zee, The Netherlands May 2026
- **Warp-induced phase space spirals** (Flash talk)
Winding, unwinding and rewinding the Gaia phase spiral (Lorentz workshop), Leiden, The Netherlands Aug 2025
- **Tilting Rate of the Milky Way Disc and Gaia Phase Spiral** (Short presentation)
Netherlands Research School for Astronomy (NOVA) PhD school, Dwingeloo, The Netherlands Mar 2025
- **Tilting rate of the Milky Way disk** (Poster with short presentation)
European Astronomical Society Annual Meeting (EAS 2024), Padova, Italy Jul 2024

SELECTED WORKSHOPS AND TRAININGS

- **XVII Scientific Meeting of the Spanish Astronomical Society**, Tarragona, Spain Jul 2026
- **EDUCADO-MWGaiaDN training school on astro-AI and ML**, Ghent, Belgium Mar 2026
- **ESR School on transferable skills: managing, entrepreneurial and IPR**, Barcelona, Spain Jun 2025
- **Galaxy modelling and galactic centre workshop**, Guildford, UK Dec 2024
- **ESR school: introduction to astro-statistics and data visualisation**, Coimbra, Portugal Sep 2024
- **The art of astrometry and computation**, Lund, Sweden Jun 2024
- **MWGaiaDN ESR induction school & diversity, public engagement and communication**, Leiden, The Netherlands Jan 2024

TEACHING EXPERIENCE

- **Astronomical Observing Techniques** (B.Sc. course), Leiden University
Teaching assistant, graded coursework, led weekly tutorial sessions (three semesters) 2024 - 2026

GRANTS AND AWARDS

- Student Scholarship at the University of Chicago (\$18 472) 2020
- Outstanding Undergraduate Thesis 2020
- Second Class of the Merit Student Scholarship for Exceptional Performance (¥3 000) 2017 & 2018

ACADEMIC SERVICE

- **Organizing committee member**, EDUCADO-MWGaiaDN training school on astro-AI and ML, Ghent, Belgium Oct 2025 - Mar 2026

REFERENCES

- **Anthony G.A. Brown**
✉ brown@strw.leidenuniv.nl
- **Victor P. Debattista**
✉ vpdebattista@gmail.com

Leiden University

University of Lancashire